

Online Event Registration System Test Design Specification Using Equivalence Partitioning (ECP) and Boundary Value Analysis (BVA)

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Project Title: Online Event Registration System (Hopeapp)

Introduction

This document presents the test design and validation process for an Online Event Registration System using Black-Box Testing techniques, specifically Equivalence Partitioning (EP) and Boundary Value Analysis (BVA). The objective of this project is to identify valid and invalid input conditions, define test scenarios, and design comprehensive test cases that verify the functionality, reliability, and robustness of the system.

The testing scope covers key modules including User Registration, Event Selection, Payment Processing, Email Confirmation, Profile Management, Password Recovery, Event Details, User Roles, Search Functionality, Reviews and Ratings, Notifications, Discounts, Session Management, Multi-Language Support, Payment Plans, Referral Programs, and other business-critical features. Through systematic partitioning of input domains and boundary testing, this project ensures that application behavior is validated against both expected and edge-case conditions, helping to improve software quality and reduce defects before deployment.

1. User Registration

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Full Name	1-50 alphabetic characters	Empty value includes numbers includes special characters more than 50 characters	1 character 50 characters	0 characters 51 characters contains numbers or special characters

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Email Address	Valid email format (e.g., user@example.com)	Empty value invalid format (e.g., missing '@', missing domain)	user@example.com	Empty missing @ missing domain contains special characters

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Age	Integer between 18 and 100	Below 18 above 100 non-integer values	18 100	17 101 non-integer values

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Password	Minimum 8 characters at least 1 uppercase letter 1 number	Less than 8 characters no uppercase letters no numbers	8 characters with uppercase and number	Less than 8 characters no uppercase no number

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Confirm Password	Matches Password field exactly	Does not match Password field empty value	Matches Password	Does not match Password empty

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Credit Card Info	16 numeric digits	Empty value non-numeric characters less than 16 digits more than 16 digits	16 digits (e.g., 1234567812345678)	Empty non-numeric 15 digits 17 digits

Test cases

Test Case	Description	Expected Outcome	New Tags Covered
1	Full Name: John Doe Email Address: john@example.com Age: 18 Password: Password1 Confirm Password: Password1	Registration successful	V1, V2, V3, V4

	Credit Card Info: 1234567812345678		
2	Full Name: Alice Email Address: alice@site.com Age: 100 Password: StrongPass9 Confirm Password: StrongPass9 Credit Card Info: 8765432187654321	Registration successful	V1, V2, V3, V4
3	Full Name: (empty) Email Address: user@domain.com Age: 25 Password: Test1234 Confirm Password: Test1234 Credit Card Info: 1234123412341234	Error: Full Name cannot be empty	I1

4	Full Name: Bob123 Email Address: bob@domain.com Age: 30 Password: MyPass123 Confirm Password: MyPass123 Credit Card Info: 4321432143214321	Error: Full Name cannot contain numbers	I2
5	Full Name: Alice! Email Address: alice!@site.com Age: 45 Password: HelloWorld9 Confirm Password: HelloWorld9 Credit Card Info: 5432543254325432	Error: Full Name cannot contain special chars	I3
6	Full Name: LongNameExceedingFiftyCh Email Address: longuser@site.com Age: 50 Password: Password1 Confirm Password: Password1 Credit Card Info: 1234123412345678	Error: Full Name exceeds 50 characters	I4
7	Full Name: John Doe Email Address: johnexample.com Age: 22	Error: Invalid email format	I5

	Password: PassWord1 Confirm Password: PassWord1 Credit Card Info: 5678567856785678		
8	Full Name: John Doe Email Address: @domain.com Age: 19 Password: PassWord1 Confirm Password: PassWord1 Credit Card Info: 6789678967896789	Error: Invalid email format	I6

9	Full Name: Jane Doe Email Address: jane@domain.com Age: 17 Password: Welcome123 Confirm Password: Welcome123 Credit Card Info: 4567456745674567	Error: Age below 18	I7
10	Full Name: John Smith Email Address: john@smith.com Age: 101 Password: SafePass45 Confirm Password: SafePass45 Credit Card Info: 2345234523452345	Error: Age above 100	I8
11	Full Name: Sara Lee Email Address: sara@domain.com Age: 25 Password: short Confirm Password: short Credit Card Info: 3456345634563456	Error: Password less than 8 characters	I9
12	Full Name: Max Payne Email Address: max@domain.com Age: 35 Password: noUppercase1 Confirm Password: noUppercase1 Credit Card Info: 1234567812345678	Error: Password must contain uppercase letters	I10
13	Full Name: Tony Stark Email Address: tony@domain.com Age: 40	Error: Password must contain numbers	I11
	Password: UPPERCASE! Confirm Password: UPPERCASE! Credit Card Info: 9876543298765432		

14	Full Name: Emma Stone Email Address: emma@domain.com Age: 28 Password: Password123 Confirm Password: Password123 Credit Card Info: 123412341234123	Error: Credit Card Info less than 16 digits	I12
15	Full Name: Rick Grimes Email Address: rick@domain.com Age: 30 Password: Zombie123 Confirm Password: Zombie123 Credit Card Info: 12345678901234567	Error: Credit Card Info more than 16 digits	I13
16	Full Name: Carol Peletier Email Address: carol@domain.com Age: 29 Password: Hello1234 Confirm Password: (empty) Credit Card Info: 2345234523452345	Error: Confirm Password cannot be empty	I14
17	Full Name: John Snow Email Address: john@snow.com Age: 30 Password: Winter123 Confirm Password: WrongMatch Credit Card Info: 3456345634563456	Error: Confirm Password does not match	I15

2. Event Selection

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Event List	Event list is populated	Empty event list	Event list with available events	No events available

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Event Selection	User selects a single event	No event selected, multiple events selected when only one is allowed	Single selected event	No selection multiple selections

Test cases

Test Case	Description	Expected Outcome	New Tags Covered
1	Event List: Populated list Event Selection: One event selected	Event details page opens for the selected event	V1, V2
2	Event List: Empty list Event Selection: -	Error: No events available	I1
3	Event List: Populated list Event Selection: No event selected	Error: Please select an event	I2
4	Event List: Populated list Event Selection: Multiple events selected	Error: Only one event can be selected at a time	I3

3. Payment Processing

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Card Number	16-digit numeric	Non-numeric characters <16 digits	1234567812345678	Non-numeric 15 digits
		>16 digits		17 digits

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Expiration Date	Future date	Past date invalid format (e.g., 13/2024)	Current year + future month/year	Past date month > 12 invalid format

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
CVV	3-digit numeric for Visa/MasterCard	Non-numeric characters <3 digits >3 digits	001 999	Non-numeric 2 digits 4 digits

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Payment Amount	Greater than 0	Zero, negative amount	Small positive value (e.g., 1)	Zero, negative value (-1)

Test cases

Test Case	Description	Expected Outcome	New Tags Covered
1	Card Number: 1234567812345678 Expiration Date: 12/2025 CVV: 123 Payment Amount: 50.00 Authorization Status: Authorized	Payment processed successfully	V1, V2, V3, V4, V5
2	Card Number: 1234abcd1234efgh	Error: Card number must be numeric	I1

	Expiration Date: 12/2025 CVV: 123 Payment Amount: 50.00 Authorization Status: Authorized		
3	Card Number: 12345678 Expiration Date: 12/2025 CVV: 123 Payment Amount: 50.00 Authorization Status: Authorized	Error: Card number must be 16 digits	I2
4	Card Number: 12345678123456781234 Expiration Date: 12/2025 CVV: 123 Payment Amount: 50.00 Authorization Status: Authorized	Error: Card number must be 16 digits	I3
5	Card Number: 1234567812345678 Expiration Date: 01/2022 CVV: 123 Payment Amount: 50.00 Authorization Status: Authorized	Error: Expiration date cannot be in the past	I4
6	Card Number: 1234567812345678 Expiration Date: invalid date CVV: 123 Payment Amount: 50.00 Authorization Status: Authorized	Error: Invalid expiration date format	I5
7	Card Number: 1234567812345678 Expiration Date: 12/2025 CVV: abc Payment Amount: 50.00 Authorization Status: Authorized	Error: CVV must be numeric	I6
8	Card Number: 1234567812345678 Expiration Date: 12/2025 CVV: 12 Payment Amount: 50.00 Authorization Status: Authorized	Error: CVV must be 3 digits	I7

9	Card Number: 1234567812345678 Expiration Date: 12/2025 CVV: 12345	Error: CVV must be 3 digits	I8
	Payment Amount: 50.00 Authorization Status: Authorized		
10	Card Number: 1234567812345678 Expiration Date: 12/2025 CVV: 123 Payment Amount: 0.00 Authorization Status: Authorized	Error: Payment amount must be greater than zero	I9
11	Card Number: 1234567812345678 Expiration Date: 12/2025 CVV: 123 Payment Amount: -10.00 Authorization Status: Authorized	Error: Payment amount cannot be negative	I10
12	Card Number: 1234567812345678 Expiration Date: 12/2025 CVV: 123 Payment Amount: 50.00 Authorization Status: Not Authorized	Error: Payment authorization failed	I11
13	Card Number: 1234567812345678 Expiration Date: 12/2025 CVV: 123 Payment Amount: 50.00 Authorization Status: Gateway Error	Error: Payment gateway encountered an error	I12

4. Email Confirmation

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Recipient Email Address	Valid email format (e.g., user@example.com)	Empty value, missing @,	user@example.com	Empty, userexample.com, user@.com
		missing domain		

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Event Details	Complete event details	Missing or incomplete details	Complete information (all details present)	Incomplete or missing details

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Email Content	Contains event name, date, time, location, and registration ID	Missing critical information (e.g., no event name or registration ID)	Content includes all required information	Missing event name, registration ID, or other required information

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Email Subject	Relevant subject (e.g., "Event Registration Confirmation")	Empty subject, unrelated subject, or no subject	"Event Registration Confirmation"	Empty, unrelated (e.g., "Hello"), or missing subject

Test cases

Test Case	Description	Expected Outcome	New Tags Covered
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1	Recipient Email: user@example.com Event Details: Complete Email Server Connectivity: Available Email Content: Full details Email Subject: "Event Registration Confirmation"	Confirmation email sent successfully	V1, V2, V3, V4, V5
2	Recipient Email: (empty) Event Details: Complete Email Server Connectivity: Available Email Content: Full details Email Subject: "Event Registration Confirmation"	Error: Recipient email cannot be empty	I1
3	Recipient Email: userexample.com Event Details: Complete Email Server Connectivity: Available Email Content: Full details Email Subject: "Event Registration Confirmation"	Error: Invalid email format	I2
4	Recipient Email: user@example.com Event Details: Incomplete Email Server Connectivity: Available Email Content: Full details Email Subject: "Event Registration Confirmation"	Error: Event details are incomplete	I3
7	Recipient Email: user@example.com Event Details: Complete Email Server Connectivity: Available Email Content: Missing event name Email Subject: "Event Registration Confirmation"	Error: Email content missing event name	I6

8	Recipient Email: user@example.com Event Details: Complete Email Server Connectivity: Available Email Content: Missing registration ID Email Subject: "Event Registration Confirmation"	Error: Email content missing registration ID	I7
9	Recipient Email: user@example.com Event Details: Complete	Error: Email subject cannot be empty	I8
	Email Server Connectivity: Available Email Content: Full details Email Subject: (empty)		
10	Recipient Email: user@example.com Event Details: Complete Email Server Connectivity: Available Email Content: Full details Email Subject: "Random Subject"	Error: Email subject is unrelated to confirmation	I9

5. Profile Management

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Username	1-50 alphanumeric characters	Empty value, includes special characters, more than 50 characters	1 character, 50 characters	0 characters, >50 characters, special characters

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
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Password	Minimum 8 characters with at least 1 uppercase letter and 1 number	Less than 8 characters, no uppercase letter, no number	8 characters with uppercase and number	Less than 8 characters, no uppercase letter, no number
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Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Full Name	1-50 alphabetic characters	Empty value, includes numbers, includes special characters, more than 50 characters	1 character, 50 characters	0 characters, 51 characters, contains numbers or special characters

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Email Address	Valid email format (e.g., user@example.com)	Empty value, invalid format (e.g., missing '@', missing domain)	user@example.com	Empty, missing @, missing domain

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Event List View	User can view registered events	No registered events	Registered events available	No events available

Test cases

Test Case	Description	Expected Outcome	New Tags Covered
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1	Username: validUser123 Password: StrongPass1	Login successful ; Profile page displays with personal info and events	V1, V2
2	Username: (empty) Password: StrongPass1	Error: Username cannot be empty	I1
3	Username: user!@# Password: StrongPass1	Error: Username cannot contain special characters	I2
4	Username: userOver50Chars_ABCDEFGHIJKLMNOPQRSTUVWXYZ_12	Error: Username cannot	I3

	3456789 Password: StrongPass1	exceed 50 characters	
5	Username: u Password: StrongPass1	Login successful ; Profile page displays with personal info and events	V3
6	Username: validUser123 Password: (empty)	Error: Password cannot be empty	I4

7	Username: validUser123 Password: weakpass	Error: Password must contain at least 1 uppercase letter and 1 number	I5
8	Username: validUser123 Password: Short1	Error: Password must be at least 8 characters	I6
9	Username: validUser123 Password: Pass1234	Error: Password must contain at least one uppercase letter	I7
10	Username: validUser123 Password: StrongPass1	Login successful ; Profile page displays with personal	V4

		info and events	
11	Field: Full Name Input Value: John Doe	Profile updated successfully	V5
12	Field: Full Name Input Value: (empty)	Error: Full Name cannot be empty	I8

13	Field: Full Name Input Value: John123	Error: Full Name cannot contain numbers	I9
14	Field: Full Name Input Value: John@Doe	Error: Full Name cannot contain special characters	I10
15	Field: Full Name Input Value: J	Profile updated successfully	V6
16	Field: Full Name Input Value: NameWithMoreThan50Chars_ABCDEFGHIJKLMNOPQRST	Error: Full Name cannot exceed 50 characters	I11
17	Field: Email Address Input Value: user@example.com	Profile updated successfully	V7
18	Field: Email Address Input Value: (empty)	Error: Email cannot be empty	I12
19	Field: Email Address Input Value: userexample.com	Error: Invalid email	I13
		format (missing '@')	
20	Field: Email Address Input Value: user@	Error: Invalid email format (missing domain)	I14

21	Event List: Registered events exist	Registered events are displayed under user profile	V8
22	Event List: No registered events	Message displayed: "No events registered"	I15

6. Password Recovery

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Email Address	Valid email format (e.g., user@example.com)	Empty value, invalid format (e.g., missing '@', missing domain)	user@example.com	Empty, missing @, missing domain

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Verification Code	6-digit numeric code	Non-numeric characters, less than 6 digits, more than 6 digits	123456	Non-numeric, 5 digits, 7 digits

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
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New Password	Minimum 8 characters with at least 1 uppercase letter and 1 number	Less than 8 characters, no uppercase letter, no number	8 characters with uppercase and number	Less than 8 characters, no uppercase letter, no number
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Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Confirm New Password	Matches New Password exactly	Does not match New Password, empty value	Matches New Password	Does not match New Password, empty

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	Email Address: user@example.com Verification Code: 123456 New Password: Password1 Confirm New Password: Password1	Password reset successful	V1, V2, V3, V4
2	Email Address: (empty) Verification Code: 123456 New Password: Password1 Confirm New Password: Password1	Error: Email Address cannot be empty	I1
3	Email Address: userexample.com Verification Code: 123456 New Password: Password1 Confirm New Password: Password1	Error: Invalid email format	I2

4	Email Address: user@ Verification Code: 123456 New Password: Password1 Confirm New Password: Password1	Error: Invalid email format	I3
5	Email Address: user@example.com Verification Code: abcdef New Password: Password1 Confirm New Password: Password1	Error: Verification Code must be numeric	I4
6	Email Address: user@example.com Verification Code: 12345 New Password: Password1 Confirm New Password: Password1	Error: Verification Code must be 6 digits	I5
7	Email Address: user@example.com Verification Code: 1234567 New Password: Password1 Confirm New Password: Password1	Error: Verification Code must be 6 digits	I6
8	Email Address: user@example.com Verification Code: 123456 New Password: Pass1 Confirm New Password: Pass1	Error: Password must be at least 8 characters	I7
9	Email Address: user@example.com Verification Code: 123456 New Password: password1 Confirm New Password: password1	Error: Password must contain an uppercase letter	I8
10	Email Address: user@example.com Verification Code: 123456 New Password: PASSWORD Confirm New Password: PASSWORD	Error: Password must contain a number	I9

11	Email Address: user@example.com Verification Code: 123456 New Password: Password1 Confirm New Password: (empty)	Error: Confirm Password cannot be empty	I10
12	Email Address: user@example.com Verification Code: 123456 New Password: Password1 Confirm New Password: Password2	Error: Confirm Password does not match New Password	I11

7. Event Details Page

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Event ID	Existing event ID	Non-existing event ID Empty ID Invalid format	Event ID exists	Non-existing ID Empty

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Event Date	Valid date format Future or current date	Past date Invalid date format	Current/future date	Past date Non-date input

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Event Time	Valid time format (HH)	Invalid time format	00:00 - 23:59	Non-time input Incorrect format

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
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Event Location	Textual location (alphanumeric with special chars)	Empty Too short (less than 2 characters)	Any nonempty location	Empty Singlecharacter location
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Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Event Description	Text description with alphanumeric and some special chars	Empty Excessive length (above 500 characters)	1-500 characters	Empty More than 500 characters

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	Event ID: 12345 Event Date: 2025-12-25 Event Time: 18:00 Event Location: Central Hall Event Description: Annual Music Concert	Event details page displays correctly with all details	V1, V2, V3, V4
2	Event ID: (empty) Event Date: 2025-12-25 Event Time: 18:00 Event Location: Central Hall Event Description: Annual Music Concert	Error: Event ID cannot be empty	I1
3	Event ID: abcde Event Date: 2025-12-25 Event Time: 18:00 Event Location: Central Hall Event Description: Annual Music Concert	Error: Invalid Event ID format	I2
4	Event ID: 99999 (non-existing) Event Date: 2025-12-25 Event Time: 18:00 Event Location: Central Hall Event Description: Annual Music Concert	Error: Event does not exist	I3

5	Event ID: 12345 Event Date: 2020-01-01 Event Time: 18:00 Event Location: Central Hall Event Description: Annual Music Concert	Error: Event Date cannot be in the past	14
6	Event ID: 12345 Event Date: invalid date Event Time: 18:00 Event Location: Central Hall Event Description: Annual Music Concert	Error: Invalid date format	15
7	Event ID: 12345 Event Date: 2025-12-25 Event Time: 24:00 Event Location: Central Hall Event Description: Annual Music Concert	Error: Invalid time format	16
8	Event ID: 12345 Event Date: 2025-12-25 Event Time: invalid time Event Location: Central Hall Event Description: Annual Music Concert	Error: Invalid time format	17
9	Event ID: 12345 Event Date: 2025-12-25 Event Time: 18:00 Event Location: (empty) Event Description: Annual Music Concert	Error: Event Location cannot be empty	18
10	Event ID: 12345 Event Date: 2025-12-25 Event Time: 18:00 Event Location: C Event Description: Annual Music Concert	Error: Event Location must be more than 1 character	19

11	Event ID: 12345 Event Date: 2025-12-25 Event Time: 18:00 Event Location: Central Hall Event Description: (empty)	Error: Event Description cannot be empty	I10
12	Event ID: 12345 Event Date: 2025-12-25 Event Time: 18:00 Event Location: Central Hall Event Description: Very long text exceeding 500 characters	Error: Event Description exceeds maximum length	I11

8. User Roles

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
User Role	Regular user Administrator	Non-existent role Empty role	Regular Administrator	Empty Undefined role

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Access to User Features	Regular user can access regular features only	Regular user accessing admin features	Regular feature access	Unauthorized access attempt by regular user

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Access to Admin Features	Administrator can access admin and regular features	Administrator restricted from own features	Admin feature access	Admin feature restriction

Test cases

Test Case	Description	Expected Outcome	Tags Covered
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1	User Role: Regular User Accessing: User Profile, Event Registration	Access granted to all user features	V1, V2
2	User Role: Regular User Accessing: Admin Dashboard	Error: Unauthorized access to admin feature	I1
3	User Role: Administrator Accessing: Admin Dashboard, User Profile	Access granted to both admin and user features	V1, V2, V3
4	User Role: Administrator Accessing: Event Management, User Management	Access granted to all adminlevel features	V1, V2, V3, V4
5	User Role: (empty) Accessing: User Profile	Error: User role cannot be empty	I2
6	User Role: Guest Accessing: User Profile	Error: Invalid role - access denied	I3
7	User Role: Regular User Accessing: Password Reset, Event Registration	Access granted to user-level features	V1, V4
8	User Role: Administrator Accessing: Restricted Admin Feature	Access granted to the restricted admin feature	V2, V3
9	User Role: Superuser Accessing: Admin Dashboard	Error: Invalid role - access denied	I4
10	User Role: Administrator Accessing: Regular User Only Feature	Error: Unauthorized feature for admin role	I5

9. Search Functionality

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Date	Format: MM/DD/YYYY Valid date	Invalid date format	Valid month Valid day Valid year	Empty date Incorrect format

		Invalid date values		
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Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Location	1-50 characters Alphanumeric with some special characters allowed	Empty value More than 50 characters	1 to 50 characters	Empty location Over 50 characters

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Category	Alphabetical characters without spaces	Numbers Spaces Special characters	Valid category (e.g., Music)	Category with numbers Spaces Special characters

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	Date: 12/25/2025 Location: Central Park Category: Music	Search results are displayed for specified filters	V1, V2, V3
2	Date: (empty) Location: Central Park Category: Music	Error: Date cannot be empty	I1
3	Date: 12-25-2025 Location: Central Park Category: Music	Error: Invalid date format (must be MM/DD/YYYY)	I2

4	Date: 25/12/2025 Location: Central Park Category: Music	Error: Invalid date format (must be MM/DD/YYYY)	I3
5	Date: 13/25/2025 (invalid month) Location: Central Park Category: Music	Error: Invalid date - month out of range	I4
6	Date: 12/25/2025 Location: (empty) Category: Music	Error: Location cannot be empty	I5
7	Date: 12/25/2025 Location: Central Parkaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaaa aaaaaa Category: Music	Error: Location exceeds maximum length of 50 characters	I6
8	Date: 12/25/2025 Location: Central Park Category: Music123	Error: Category can only contain alphabets	I7
9	Date: 12/25/2025 Location: Central Park Category: Music Concert	Error: Category cannot contain spaces	I8
10	Date: 12/25/2025 Location: Central Park Category: Art	Search results are displayed for specified filters	V1, V2, V3
11	Date: 01/01/2024 Location: Madison Square Garden Category: Sports	Search results are displayed for specified filters	V1, V2, V3

12	Date: 02/29/2024 (leap year) Location: Main Hall Category: Science	Search results are displayed for specified filters	V1, V2, V3
13	Date: 12/25/2025 Location: Park 52 Category: Film	Search results are displayed for specified filters	V1, V2, V3

10. User Reviews and Ratings

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Rating (Stars)	1 to 5	Less than 1	1	0
		Greater than 5	5	6
		Non-numeric		

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Review	0 to 200 characters	More than 200 characters	1 character	201 characters
		Non-alphanumeric characters	200 characters	Non-numeric

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User enters Rating: 5 Review: "Great experience!"	Rating and review submitted successfully Displays "5 - great experience"	V1, V2, V3, V4, V5

2	User enters Rating: 1 Review: ""	Rating submitted successfully with no review Displays "1 - very bad experience"	B1, B3, B5
3	User enters Rating: 3 Review: "It was fine nothing special."	Rating and review submitted successfully Displays "3 - fine"	V1, V3
4	User enters Rating: 6 Review: "Exceeded expectations."	Error message: "Invalid rating, must be between 1 and 5"	I1, I2
5	User enters Rating: 4 Review: [201-character review]	Error message: "Review must be 200 characters or less"	I3
6	User enters Rating: "bad" Review: "Loved it!"	Error message: "Invalid rating, please enter a number between 1 and 5"	I4
7	User enters Rating: 2 Review: "Decent."	Rating and review submitted successfully Displays "2 - bad experience"	V1, V2
8	User enters Rating: 0 Review: "Terrible event!"	Error message: "Invalid rating, must be between 1 and 5"	I1

11. Notification System

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Notification Type	Event Reminder Event Details Change General Updates	Undefined Type Empty Type	N/A	Non-standard types

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Delivery Time	At event creation Before event start time Real-time	After event end time Null value	Before event time	After event ends

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	Notification Type: Event Reminder Delivery Time: Before event start time	Notification sent successfully before event starts Displays event reminder notification	V1, V2, V3
2	Notification Type: Event Details Change Delivery Time: Realtime	Notification sent immediately Displays "Event details have changed" notification	V1, V2, V3
3	Notification Type: General Updates Delivery Time: At event creation	Notification sent successfully upon event creation Displays general update notification	V1, V2, V3
4	Notification Type: Undefined Type Delivery Time: Realtime	Error message: "Invalid notification type"	I1
5	Notification Type: Event Reminder Delivery Time: After event end time	Error message: "Notification cannot be sent after the event has ended"	I2
6	Notification Type: Event Reminder Delivery Time: Null value	Error message: "Notification delivery time cannot be empty"	I3
7	Notification Type: Empty Type Delivery Time: Before event start time	Error message: "Notification type cannot be empty"	I4

12. Event Cancellation

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
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Cancellation Time Frame	Before 10 days of the event Before 5 days of the event After 5 days of the event	After event start Null value	10 days before 5 days before	0 days before After event starts
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Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Refund Amount	100% refund 50% refund 0% refund	Negative refund Greater than 100%	100% refund 50% refund 0% refund	More than 100% Less than 0%

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User cancels registration 12 days before the event	Full refund issued (100% of the amount)	V1, V2
2	User cancels registration 8 days before the event	Full refund issued (100% of the amount)	V1, V2
3	User cancels registration 6 days before the event	50% refund issued	V1, V2
4	User cancels registration 3 days before the event	No refund issued (0% of the amount)	V1, V3
5	User attempts cancellation after the event starts	Error message: "Cancellation not allowed after event has started"	I1
6	User cancels registration with null time frame	Error message: "Cancellation time frame cannot be empty"	I2
7	User cancels registration 10 days before the event	Full refund issued (100% of the amount)	B1
8	User cancels registration exactly 5 days before the event	50% refund issued	B2
9	User cancels registration 1 day before the event	No refund issued (0% of the amount)	V1, V3

13. Promotional Discount Codes

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Discount Code Format	4-digit numeric code	Less than 4 digits	1000	999
		More than 4 digits	9999	10000
		Non-numeric characters		
		Null value		

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User enters discount code: 1234	Discount code applied successfully	V1, V2
2	User enters discount code: 0000	Discount code applied successfully	V1, V3
3	User enters discount code: 9999	Discount code applied successfully	V1, V3
4	User enters discount code: 123	Error message: "Discount code must be a 4-digit number"	I1
5	User enters discount code: 12345	Error message: "Discount code must be a 4-digit number"	I2
6	User enters discount code: ABCD	Error message: "Discount code must contain only numbers"	I3
7	User enters discount code: 12AB	Error message: "Discount code must contain only numbers"	I4
8	User enters discount code: (empty)	Error message: "Discount code cannot be empty"	I5

14. Mobile Responsiveness

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Screen Size	Mobile screen size ($\leq 768\text{px}$)	Extremely small (less than 320px)	320px	Less than 320px
	Tablet screen size (768-1024px)	Extremely large (above 2000px)	1024px	More than 2000px
	Desktop screen size ($>1024\text{px}$)			

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	Open system on a mobile device (width: 375px)	UI adjusts to mobile layout; all elements are accessible and well-positioned	V1, V2
2	Open system on a tablet device (width: 800px)	UI adjusts to tablet layout; all elements are accessible and well-positioned	V1, V3
3	Open system on a desktop device (width: 1440px)	UI adjusts to desktop layout; all elements are accessible and well-positioned	V1, V4
4	Open system on a screen with width less than 320px	Error or message prompting the user to use a supported screen size	I1
5	Open system on a screen with width greater than 2000px	UI may not fully adjust; message suggesting optimal resolution for best view	I2

15. Data Export

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Export Format	CSV Excel	Unsupported format Null	CSV	XML, JSON

			Excel	
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Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	Admin exports data in CSV format	Data exported successfully in CSV format	V1
2	Admin exports data in Excel format	Data exported successfully in Excel format	V2
3	Admin attempts to export data in XML format	Error message: "Unsupported file format"	I1
4	Admin attempts to export data in JSON format	Error message: "Unsupported file format"	I2
5	Admin attempts export without selecting any format	Error message: "Please select an export format"	I3

16. Session Management

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Inactivity Timeout Duration	10 - 30 minutes	Less than 1 minute	10 minutes	Less than 1 minute
		More than 60 minutes	30 minutes	More than 60 minutes

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User remains inactive for 10 minutes	User session remains active	V1
2	User remains inactive for 30 minutes	User session remains active	V2

3	User remains inactive for 31 minutes	User session expires; user is logged out	B1
4	User remains inactive for more than 60 minutes	User session expires; user is logged out	I1
5	User remains inactive for less than 1 minute	User session remains active	I2

17. Multi-Language Support

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Language Selection	English Spanish French	Unsupported language Null	English	Unavailable language

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User selects English as the language	System switches to English successfully	V1
2	User selects Spanish as the language	System switches to Spanish successfully	V2
3	User selects French as the language	System switches to French successfully	V3
4	User selects an unsupported language (e.g., German)	Error message: "Selected language not supported"	I1
5	User does not select any language (null)	Default language (English) is applied	I2

18. Group Registration Discounts

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
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Number of Participants	1-5 participants (0% discount) 6-10 participants (10% discount) 11+ participants (15% discount)	0 participants Negative number Non-integer value	5 participants 6 participants 10 participants 11 participants	Less than 1 Non-integer
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Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User registers 3 participants	No discount applied; total cost calculated with 0% discount	V1
2	User registers 6 participants	10% discount applied; total cost calculated with 10% discount	V2
3	User registers 10 participants	10% discount applied; total cost calculated with 10% discount	V2
4	User registers 11 participants	15% discount applied; total cost calculated with 15% discount	V3
5	User registers 1 participant	No discount applied; total cost calculated with 0% discount	B1
6	User registers 5 participants	No discount applied; total cost calculated with 0% discount	B2
7	User registers 0 participants	Error message: "At least one participant is required to register"	I1
8	User registers a negative number of participants (-2)	Error message: "Invalid number of participants"	I2
9	User registers a non-integer value (e.g., 3.5)	Error message: "Number of participants must be an integer"	I3

19. Early Bird Registration

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
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Days Before Event for Registration	30 days or more (20% discount) Less than 30 days (no discount)	Negative days 0 days	30 days 29 days	Less than 0 days
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Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User registers 45 days before the event	20% discount applied; total cost calculated with 20% discount	V1
2	User registers exactly 30 days before the event	20% discount applied; total cost calculated with 20% discount	B1
3	User registers 29 days before the event	No discount applied; total cost calculated with 0% discount	B2
4	User registers 15 days before the event	No discount applied; total cost calculated with 0% discount	V2
5	User registers on the event day (0 days before the event)	No discount applied; total cost calculated with 0% discount	I1
6	User registers with a negative value for days (-10 days)	Error message: "Registration date cannot be after the event date"	I2

20. Promotional Code Validation

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Promotional Code Validity	Valid code Expired code Code used	Code not found in database	Expiry date Usage limit	Null code Expiry exceeded
	within usage limit	Code exceeded usage limit		

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Discount Percentage	Specified discount (e.g., 10%, 20%)	Discount greater than 100% Negative discount	0% 100%	Greater than 100% Less than 0%

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User enters a valid promotional code	Specified discount applied; total cost calculated with the promotional discount	V1
2	User enters an expired promotional code	Error message: "Promotional code has expired"	I1
3	User enters a code that is not found in the database	Error message: "Invalid promotional code"	I2
4	User enters a valid code but exceeds usage limit	Error message: "Promotional code usage limit exceeded"	I3
5	User enters a valid code with a 10% discount	10% discount applied; total cost calculated with 10% discount	V2
6	User enters a valid code with a 20% discount	20% discount applied; total cost calculated with 20% discount	V3
7	User enters a null or empty promotional code	Error message: "Please enter a promotional code"	I4
8	User enters a valid code with an unusually high discount (110%)	Error message: "Invalid discount percentage"	I5
9	User enters a valid code with a negative discount (10%)	Error message: "Invalid discount percentage"	I6

21. Payment Plan Options

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Payment Plan Selection	Full payment upfront Two installments Three installments	Null selection Non-existent plan	Full payment Two installments Three installments	Other payment plan Null

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Payment Amount Calculation	Correctly calculated amounts based on selected plan	Incorrect calculation Negative amount	100% upfront 50%-50% installments 33%-33%-34% installments	More than total Less than required

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User selects full payment upfront	Total amount due displayed as a single payment	V1
2	User selects two installments for a \$1000 event	First installment: \$500 due now; Second installment: \$500 due in 30 days	V2
3	User selects three installments for a \$1000 event	First installment: \$333 due now; Second installment: \$333 in 30 days; Final installment: \$334 in 60 days	V3
4	User selects an invalid payment plan option	Error message: "Invalid payment plan selection"	I1
5	User does not select any payment plan (null selection)	Error message: "Please select a payment plan"	I2
6	User selects two installments but system displays incorrect amount (e.g., \$600 and \$400)	Error message: "Payment plan amounts calculated incorrectly"	I3
7	User selects three installments with negative installment amounts	Error message: "Invalid payment amount"	I4

8	User selects three installments but receives only one due date	Error message: "Installment dates not calculated correctly"	I5
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22. Event Bundle Registration

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Number of Events in Bundle	2 events (5% discount) 3 events (10% discount) 4 or more events (15% discount)	0 events 1 event Negative number	2 events 3 events 4 events	Less than 2 events

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Discount Percentage	5% discount 10% discount 15% discount	Discount greater than 15% Negative discount	5% 10% 15%	Greater than 15% Less than 0%

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User registers for 2 events	5% discount applied; total cost calculated with 5% discount	V1, V2
2	User registers for 3 events	10% discount applied; total cost calculated with 10% discount	V1, V3
3	User registers for 4 events	15% discount applied; total cost calculated with 15% discount	V1, V4
4	User registers for 1 event	No discount applied; total cost calculated with 0% discount	I1

5	User registers for 0 events	Error message: "Please select at least 2 events to qualify for a bundle discount"	I2
6	User registers for a negative number of events (-3)	Error message: "Invalid number of events"	I3
7	User registers for 2 events but system applies incorrect discount (e.g., 10%)	Error message: "Discount applied incorrectly"	I4
8	User registers for 4 events but system applies discount greater than 15%	Error message: "Invalid discount percentage"	I5

23. Customizable Registration Options

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Registration Option Selection	Main event only Main event + workshop Main event + meal preference	Null selection Invalid option	Main event + workshop	Other invalid combinations

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Discount Trigger	Workshop selected with main event (10% discount on workshop)	Workshop only Invalid combination	Main event + workshop	Non-triggering combinations

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User selects main event only	No additional discount applied; total cost calculated without discounts	V1

2	User selects main event + workshop	10% discount applied on the workshop fee; total cost calculated with discount	V2
3	User selects main event + meal preference	No discount applied; total cost calculated with full price for both selections	V3
4	User selects main event, workshop, and meal preference	10% discount applied on the workshop fee; total cost calculated with discount	V2, V3
5	User selects workshop only without main event	No discount applied; error message: "Workshop can only be selected with main event"	I1
6	User does not select any registration option (null)	Error message: "Please select at least one registration option"	I2
7	User selects an invalid option (e.g., parking)	Error message: "Invalid option selected"	I3
8	User selects main event + workshop but receives no discount	Error message: "Discount not applied correctly"	I4
9	User selects main event + workshop, system applies greater than 10% discount	Error message: "Invalid discount percentage"	I5

24. Referral Program

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Referral Code Validity	Valid code New user registers with referral	Code not found in database Expired code	Any valid code	Null code Invalid characters

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Discount Application	\$10 discount for both users when referral code is used	Incorrect discount amount No	\$10 discount	Greater than \$10 Less than \$10

		discount applied		
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Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	New user registers with a valid referral code	\$10 discount applied to both referrer and new user on their next registration	V1, V2
2	New user registers without entering a referral code	No discount applied; standard registration process completes	I1
3	New user registers with an invalid referral code	Error message: "Invalid referral code"	I2
4	New user registers with an expired referral code	Error message: "Referral code has expired"	I3
5	New user registers with a valid code, but receives incorrect discount (e.g., \$15)	Error message: "Incorrect discount amount applied"	I4
6	New user registers with a valid referral code, but no discount is applied	Error message: "Discount not applied"	I5
7	User enters a referral code containing invalid characters (e.g., special symbols)	Error message: "Referral code contains invalid characters"	I6
8	User enters a null referral code	Error message: "Please enter a valid referral code"	I7

25. Age-Based Discounts

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
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Age Group	Students (ages 18-25) Seniors (ages 60+) Regular (ages 26-59, no discount)	Under 18 Age entered as a negative number Null age entry	18 years 25 years 60 years	Age less than 0 Non-integer age
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Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Registration Type	Individual registration (eligible for age discount) Group registration (not eligible for age discount)	Null registration type Invalid type selection	Individual registration	Group registration with discount applied

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User registers individually, age 20	15% discount applied for student category	V1, V2
2	User registers individually, age 62	20% discount applied for senior category	V1, V3
3	User registers individually, age 30	No discount applied; user falls within the regular age group	V1, V4
4	User registers in a group, age 20	No discount applied; group registrations are not eligible for age-based discounts	I1
5	User registers individually, age 17	Error message: "Age-based discount is only available for ages 18 and above"	I2
6	User registers individually, age 25	15% discount applied for student category	B1
7	User registers individually, age 60	20% discount applied for senior category	B2
8	User registers individually with a negative age (-5)	Error message: "Invalid age entered"	I3

9	User registers individually with age input as a noninteger (e.g., "twenty")	Error message: "Please enter a valid numerical age"	I4
10	User registers without entering an age	Error message: "Age is required for age-based discount eligibility"	I5

26. Feedback Incentives

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Feedback Submission Status	Feedback submitted Feedback not submitted	Null feedback submission	Feedback submitted	Null submission

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Discount Application	10% discount on next registration after feedback submission	No discount applied despite feedback Incorrect discount amount	10% discount	Greater than 10% Less than 10%

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User completes feedback survey after an event	10% discount applied on next event registration	V1, V2
2	User does not complete feedback survey	No discount applied; regular registration process continues	V3
3	User submits feedback, but system does not apply discount at checkout	Error message: "Discount not applied despite feedback submission"	I1

4	User submits feedback, but system applies an incorrect discount (e.g., 15%)	Error message: "Incorrect discount amount applied"	I2
5	User attempts to submit feedback but leaves the feedback form blank	Error message: "Please complete feedback survey to earn discount"	I3
6	User submits feedback, discount is correctly displayed as 10%	Discount correctly applied at checkout; total cost calculated with 10% discount	V1, V2
7	User submits feedback but does not use discount on next event	Discount remains in the system, valid for future use	V4
8	User submits feedback multiple times to receive multiple discounts	Only one 10% discount is applied to the next registration; subsequent feedback does not stack discounts	I4

27. Multiple Payment Methods

Equivalent classes and BVA

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Payment Method Selection	Credit card PayPal Bank transfer	Invalid payment method Null payment method	PayPal selected	Unsupported method

Conditions	Valid Partitions	Invalid Partitions	Valid Boundaries	Invalid Boundaries
Processing Fee Application	3% fee for PayPal No fee for other methods	Incorrect fee amount applied Fee applied to non-PayPal methods	3% of total amount for PayPal	Fee greater than or less than 3%

Test cases

Test Case	Description	Expected Outcome	Tags Covered
1	User selects credit card as payment method	No additional fee applied; total amount displayed accurately	V1, V2
2	User selects PayPal as payment method	3% processing fee added; total amount displayed accurately with PayPal fee	V1, V3
3	User selects bank transfer as payment method	No additional fee applied; total amount displayed accurately	V1, V4
4	User selects an invalid payment method (e.g., cryptocurrency)	Error message: "Invalid payment method selected"	I1
5	User selects PayPal, but system applies incorrect fee (e.g., 5%)	Error message: "Incorrect processing fee applied"	I2
6	User selects PayPal, system does not apply processing fee	Error message: "Processing fee not applied for selected method"	I3
7	User selects credit card, but system applies a processing fee	Error message: "No additional fee should be applied for credit card payments"	I4
8	User selects PayPal and sees final amount, including 3% fee	Final amount displayed accurately, with 3% PayPal fee added to the total	V1, V3
9	User does not select any payment method (null)	Error message: "Please select a payment method to proceed"	I5